

Business Travel CO₂ Dashboard

Tactical decision support for travel and sustainability managers.

Monitor budget compliance · Identify reduction levers · Plan smarter trips.

4

Business Units

4

Transport Modes

t

CO₂ Tracked

The Challenge

Corporate travel is a significant but manageable source of emissions.



No Visibility

Travel managers lack real-time CO₂ tracking across business units and routes.



Budget Drift

Without dashboards, BU carbon budgets are exceeded before anyone notices.



Missed Savings

Many short-haul flights could switch to train but no tool surfaces these opportunities.



The dashboard puts budget compliance, geographic heatmaps, and mode-shift recommendations in one place — so every decision saves both money and carbon.

Two Personas, One Tool

Anna Meier

Sustainability Manager

Goal: Monitor CO₂ budgets across all BUs and report to leadership monthly.

Pain: Data scattered across Excel files, no live overview of compliance.

Dashboard value: Real-time gauges + budget status at a glance.

Daniel Schmid

Travel Manager

Goal: Approve upcoming trips while staying within CO₂ and cost limits.

Pain: No way to see if a planned flight has a viable train alternative.

Dashboard value: Route deep-dive + one-click mode-shift alternatives.

Four Sections, One View



Overview KPIs

01

Total CO₂ vs. combined budget, utilisation %, reduction potential, and average kg per trip — at a glance.



BU Compliance

02

Gauge dials and stacked bar chart showing each business unit's actual vs. optimised CO₂ against its annual budget.



Geographic Map

03

World connection map with route lines scaled by CO₂. Filter by region; hover for per-route details.



Reduction Levers

04

Ranked table of flights with greener alternatives, route deep-dive with CO₂ and travel-time comparison, and one-click 'Apply alternatives' scenario.

Know Your Numbers Instantly

847 t

Total CO₂
+12% vs budget

112%

Budget Used
Combined BUs

-143 t

Saving Potential
via mode shift

1,118

Trips Analysed
avg 758 kg/trip

106%

BU1 190 t / 180 t
▲ 10 t over

75%

BU2 150 t / 200 t
✓ 50 t left

142%

BU3 249 t / 175 t
▲ 74 t over

98%

BU4 186 t / 190 t
~ 4 t left

BU Snapshot

BU1

Actual: 190 t · Optimised: 162 t · Budget: 180 t

▲ 10 t over

BU2

Actual: 150 t · Optimised: 134 t · Budget: 200 t

✓ 50 t left

BU3

Actual: 249 t · Optimised: 198 t · Budget: 175 t

▲ 74 t over

BU4

Actual: 186 t · Optimised: 168 t · Budget: 190 t

✓ 4 t left

Why These Visual Choices?

Gauges

Encode status (under / approaching / over) at a glance, no numbers needed to detect overruns.

Red = over budget

Amber = approaching

Green = on track

Connection Map

Shows geographic concentration of emissions. Line thickness = CO₂ volume. Color = transport mode.

Answers: Where do we fly most?

Which routes drive the most CO₂?

Ranked Table

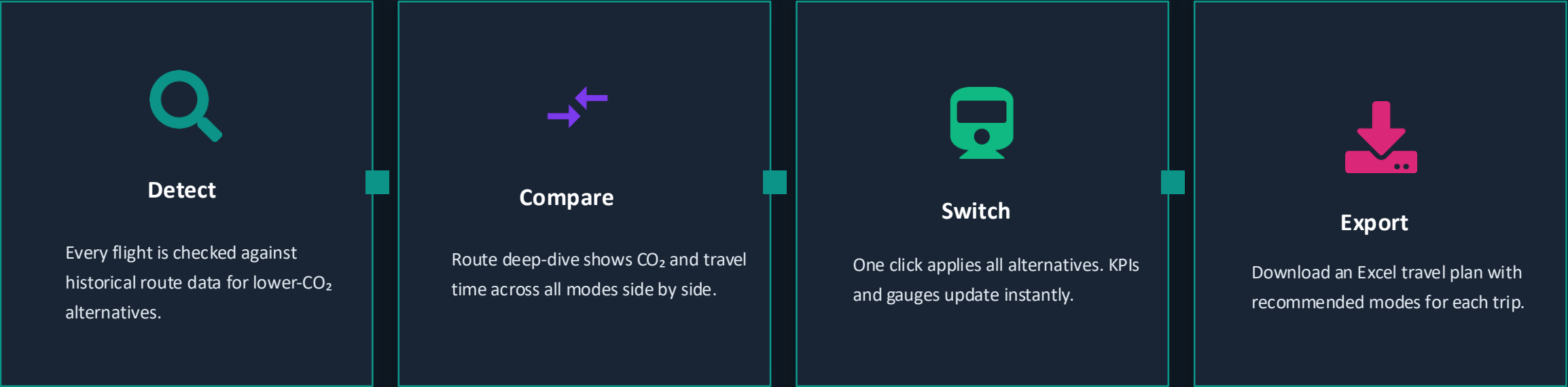
Prioritises action: highest saving potential shown first. One row = one decision a manager can act on today.

Consistent BU color palette carried across all charts for zero re-learning.

Narrative Flow

Overview → BU detail → Geography → Action. Each section answers one question before handing off to the next.

From Flights to Trains



Top Saving Routes (example)

From	To	Alternative	Flights	Saving (t)	Saving %
ZRH	AMS	Train	54	5.22	95.5%
BER	ZRH	Train	46	5.00	96.0%
ZRH	VIE	Train	54	4.85	95.6%
LJU	ZRH	Train	53	4.82	97.0%
CDG	ZRH	Train	58	4.52	95.5%

Getting Started in 3 Steps

Upload Historical Reference

Excel file with a travel_data sheet (routes, CO₂, modes) and a budget_2026 sheet mapping BU budgets in tonnes.

Upload Planned Trips

Excel with upcoming trips. Without it, the dashboard analyses the historical data itself, useful for retrospective reviews.

Select CO₂ Metric & Explore

Choose between RFI 2 or RFI 2.7 accounting. All charts, gauges and saving calculations update live.

Required Excel Schema

travel_data sheet

- ▶ transport_mode
- ▶ departure_iata / arrival_iata
- ▶ departure_lat/lon / arrival_lat/lon
- ▶ business_unit
- ▶ CO₂e RFI2 (t) or RFI2.7 (t)
- ▶ km · cost_CHF · date (optional)

budget_2026 sheet

- ▶ Business Unit (BU1, BU2, ...)
- ▶ Budget 2026 (column name must contain '2026')



Use the sidebar to switch CO₂ metric at any time, no reload needed.

Ready to Reduce Your Carbon Footprint?

- ✓ Real-time budget compliance across all BUs
- ✓ Automatic detection of mode-shift opportunities
- ✓ Interactive maps and route deep-dives
- ✓ Excel export with per-trip recommendations
- ✓ Runs locally



Get Started

```
streamlit run app.py
```